

OLIST E-COMMERCE | Logistics & Growth Strategy

Operational Performance, Delivery Bottlenecks & Customer Experience Analysis

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The Business Challenge

OBJECTIVE

Identify operational bottlenecks and customer friction points to safeguard brand reputation and drive long-term GMV growth.

1. Commercial Operations

Which product categories drive total order volume, and where is revenue concentrated?

2. Customer Experience

How severely do shipping delays damage review scores and overall customer satisfaction?

3. Logistics & Geography

Where are regional bottlenecks located, and which states suffer the highest delay rates?

The Goal: Deliver actionable, data-backed logistics strategies to reduce delivery lead times and improve overall customer review ratings.

Data & Methodology

End-to-end data pipeline from raw dataset cleaning to executive visual reporting

STEP 01: PYTHON

Data Cleaning and Prep

Cleaned raw e-commerce CSV files using Python (Pandas). Handled missing values, formatted timestamps, corrected data types, and validated table integrity

STEP 02: SQL

Relational Modeling & EDA

SQLite relational database. Joined core entities (orders, items, products, customers, reviews) and executed SQL queries to calculate lead times and delay flags

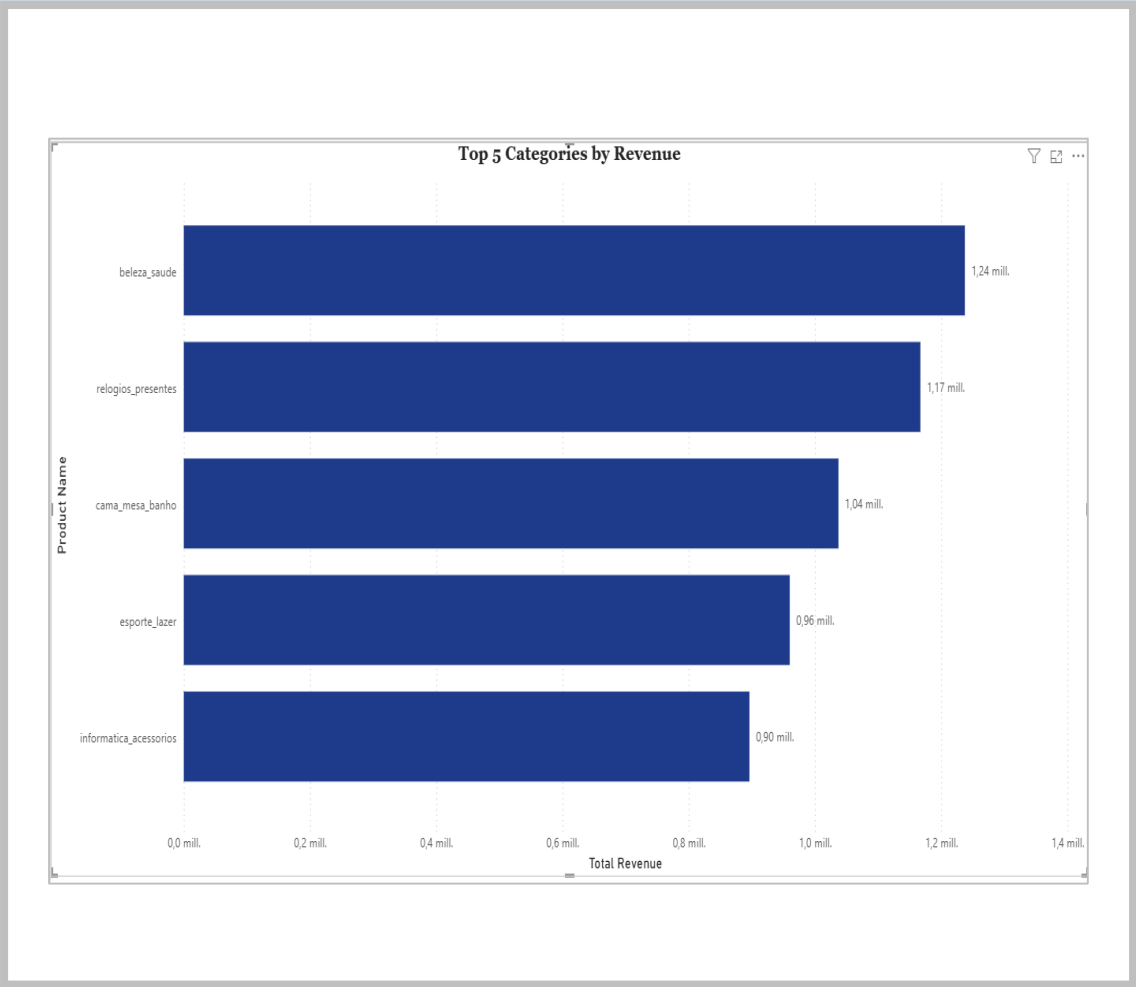
STEP 03: POWER BI

Executive Dashboard

Built dynamic DAX measures (Total Revenue, Order Volume, Average Rating, Late Delivery Rate) and created interactive visuals to communicate insights to stakeholders.

Insight 1: Commercial Performance

Health & Beauty leads revenue, while high-ticket categories drive transaction value.



Volume Leaders
beleza_saude (Health & Beauty) dominates total GMV at **\$1.24M**, supported by high daily transaction volume across all regions.

High-Value Categories
relorios_presentes (Watches & Gifts) ranks second at **\$1.17M** despite lower overall order volume, highlighting a significantly higher Average Order Value.

Top 5 categories account for a substantial share of total \$13.28M platform revenue.

Insight 2: Customer Satisfaction Impact

Shipping delays destroy customer loyalty, causing a steep drop in review ratings.

ON-TIME DELIVERIES

4.21 ★

Average Review Score

On-time orders maintain high customer satisfaction, proving product quality meets market expectations.

DELAYED DELIVERIES

2.55 ★

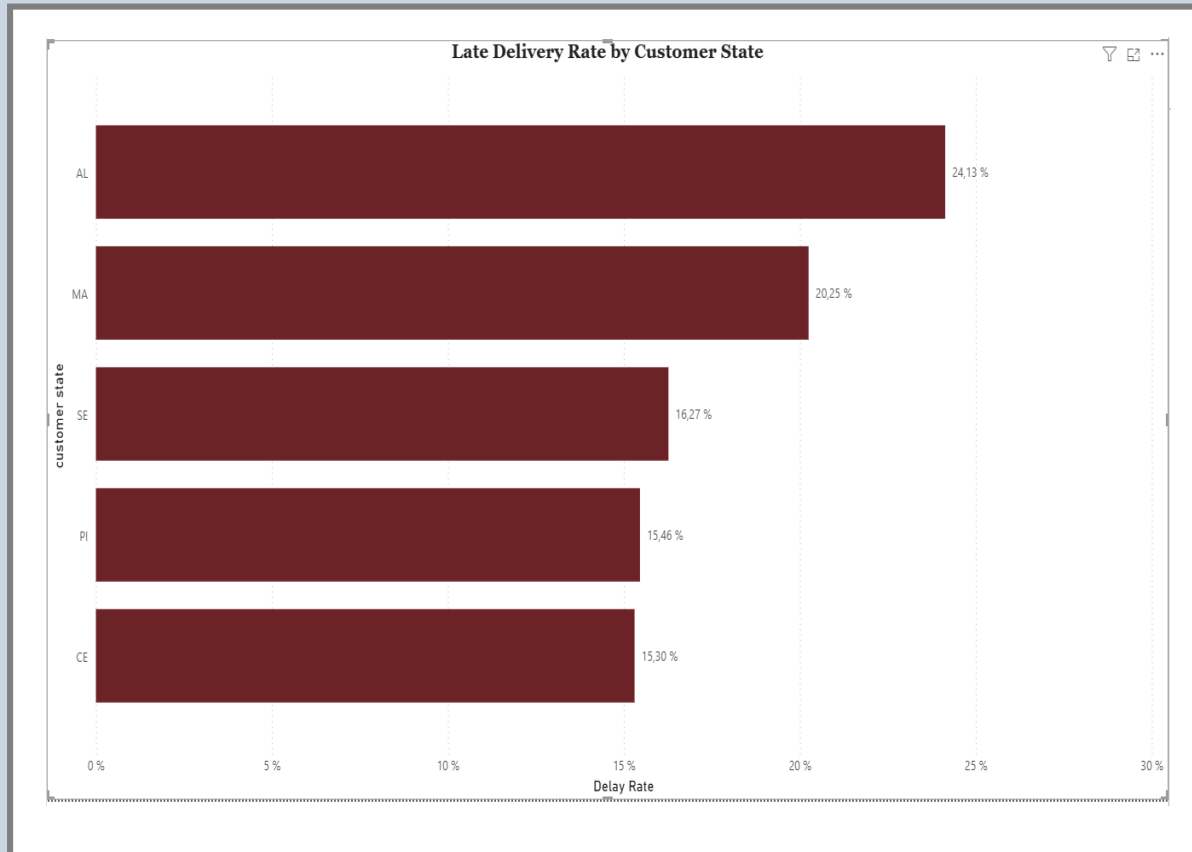
Average Review Score

A single late delivery triggers an immediate -1.66 star drop, degrading LTV and increasing customer support operational costs.

Key Finding: Product quality is not the issue—operational delivery delays are driving customer churn and negative reviews.

Insight 3: Geographic Logistics Bottlenecks

Delivery delays are heavily concentrated in remote states, severely impacting overall SLA.



Northern & Northeast Region

States like **RR, AP, and AM** face severe logistical bottlenecks, reaching delay rates above **15% - 20%** due to long-haul transit dependencies.

Southeast Core (SP, RJ, MG)

Accounts for over **60% of total order volume**. While delay percentages are lower, absolute delayed volume remains concentrated here due to heavy hub congestion.

Root Cause: A "one-size-fits-all" fulfillment network fails to address regional infrastructure disparity.

Financial Impact: Cost of Logistics Delays

Quantifying the revenue, customer satisfaction, and operational risk of unfulfilled SLAs.

REVENUE EXPOSURE

\$1.07M

GMV in Delayed Orders

Represents 8.1% of total platform GMV (\$13.28M) directly exposed to order cancellations, customer refund requests, and chargebacks.

CUSTOMER SATISFACTION

-1.66 ★

Average Rating Collapse

Delayed deliveries trigger a steep rating drop from **4.21★ to 2.55★**, generating **5.8x more 1-star reviews** and destroying customer retention.

OPERATIONAL OVERHEAD

3x - 5x

Support Ticket Surge

Unfulfilled delivery promises multiply customer support inquiries (WISMO tickets) and claims, driving up post-sale operational costs.

Strategic Action Plan & Financial ROI

Targeted operational interventions to protect GMV, restore customer ratings, and optimize SLAs.

PILLAR 1 | LOGISTICS

Local 3PL Partnerships

Partner with regional carriers in high-delay states (**AL, MA, SE**) to bypass main hub congestion and reduce long-haul transit times by **15%–20%**.

🎯 **Target ROI:** Protects +\$250K+ in regional GMV.

PILLAR 2 | OPERATIONS

Dispatch Scorecards & Penalties

Enforce automated dispatch alerts and seller scorecards. Penalize chronic late-shipping sellers to eliminate fulfillment bottlenecks before transit.

🎯 **Target ROI:** Reduces WISMO support tickets by **30%** & restores overall rating to **>4.0★**.

PILLAR 3 | CUSTOMER EXP

Dynamic Checkout Delivery Estimates

Implement dynamic delivery calculations at checkout based on real-time regional performance instead of flat estimates.

🎯 **Target ROI:** Prevents **1-star reviews** driven by unfulfilled delivery promises.

Thank You!

Questions & Discussion

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Olist E-Commerce Logistics Analysis